

Job No.	Sheet No.	Rev
exp no 08	1	
Part	Ref	
By V.UMA SRI	Date 11-Sep-26	Chd
Date Time 15-Sep-2026 14:45		

Job Title ANALYSIS AND DESIGN PORTAL FRAME WITH SWAY

Client

File ANALYSIS AND DESIGN PORTAL FRAME WITH SWAY.std

Job Information

	Engineer	Checked	Approved
Name:	V.UMA SRI		
Date:	11-Sep-26		

Comments:	
Structure Type:	SPACE FRAME

Geometry

Entity Type	Count	Highest
Nodes	4	4
Analytical Members	3	3

Load Cases

Load Case Type	Count
Primary	2
Combination	1

Included in this printout are data for:

All	The Whole Structure
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Load Case Table

Included in this printout are results for load cases:

L/C	Type	Name
1	Primary	DEAD LOAD
2	Primary	LIVE LOAD
3	Combination	COMBINATION LOAD CASE 3

Nodes

Node	X (m)	Y (m)	Z (m)
1	0.000	0.000	0.000
2	5.000	0.000	0.000
3	0.000	4.000	0.000
4	5.000	4.000	0.000

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Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	1	3	4.000	1	0.000
2	3	4	5.000	1	0.000
3	4	2	4.000	1	0.000

Beams

Beam	Node A	Node B	Length (m)	Property	β (degrees)
1	1	3	4.000	1	0.000
2	3	4	5.000	1	0.000
3	4	2	4.000	1	0.000

Supports

Node	X (kip/in)	Y (kip/in)	Z (kip/in)	rX (kip-ft/deg)	rY (kip-ft/deg)	rZ (kip-ft/deg)
1	Fixed	Fixed	Fixed	Fixed	Fixed	Fixed
2	Fixed	Fixed	Fixed	-	-	-

Combination Load Cases

Comb.	Combination L/C Name	Primary	Primary L/C Name	Factor
3	COMBINATION LOAD CASE 3	1	DEAD LOAD	1.500
		2	LIVE LOAD	1.500

Beam Displacement Summary

Type	Beam	L/C	d (m)	X (in)	Y (in)	Z (in)	Result ant (in)
Max X	3	3	1.000	0.135	-0.004	-1.830	1.835
Min X	1	1	2.000	0.000	0.000	0.000	0.000
Max Y	3	3	4.000	0.000	0.000	0.000	0.000
Min Y	3	3	0.000	0.130	-0.005	-2.350	2.354
Max Z	3	3	4.000	0.000	0.000	0.000	0.000
Min Z	3	3	0.000	0.130	-0.005	-2.350	2.354
Max Rst	2	3	5.000	0.130	-0.005	-2.350	2.354

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ANALYSIS AND DESIGN PORTAL FRAME WITH SWAY		

Beam End Forces

Beam	Node	L/C	Axial			Shear			Torsion		Bending	
			Fx	Fy	Fz	Mx	My	Mz				
			(kN)	(kN)	(kN)	(kip-in)	(kip-in)	(kip-in)				
1	1	1	27.328	-1.606	0.000	0.000	0.000	-10.588				
	3	1	-10.363	1.606	0.000	0.000	0.000	-46.252				
	1	2	40.065	33.503	-9.596	637.409	0.000	622.448				
	3	2	-40.065	-3.503	9.596	106.365	339.744	32.618				
	1	3	101.090	47.846	-14.395	956.114	0.000	917.789				
	3	3	-75.643	-2.846	14.395	159.547	509.616	-20.451				
2	3	1	1.606	10.363	0.000	0.000	0.000	46.252				
	4	1	-1.606	10.842	0.000	0.000	0.000	-56.840				
	3	2	-3.503	40.065	-9.596	339.744	106.365	-32.618				
	4	2	3.503	59.935	9.596	339.744	531.045	407.026				
	3	3	-2.846	75.643	-14.395	509.616	159.547	20.451				
	4	3	2.846	106.165	14.395	509.616	796.567	695.800				
3	4	1	10.842	-1.606	0.000	0.000	0.000	56.840				
	2	1	-27.806	-1.606	0.000	0.000	0.000	0.000				
	4	2	59.935	-3.503	-9.596	531.045	339.744	407.026				
	2	2	-59.935	-26.497	9.596	0.000	0.000	0.000				
	4	3	106.165	-2.846	-14.395	796.567	509.616	695.800				
	2	3	-131.611	-42.154	14.395	0.000	0.000	0.000				

Beam Force Detail Summary

	Beam	L/C	d	Axial			Shear			Torsion		Bending	
				Fx	Fy	Fz	Mx	My	Mz				
			(m)	(kN)	(kN)	(kN)	(kip-in)	(kip-in)	(kip-in)				
Max Fx	3	3	4.000	131.611	42.154	-14.395	0.000	0.000	0.000				
Min Fx	2	2	0.000	-3.503	40.065	-9.596	339.744	-106.365	-32.618				
Max Fy	2	3	0.000	-2.846	75.643	-14.395	509.616	-159.547	20.451				
Min Fy	2	3	5.000	-2.846	-106.165	-14.395	509.616	-796.567	695.800				
Max Fz	1	1	0.000	27.328	-1.606	0.000	0.000	0.000	-10.588				
Min Fz	1	3	0.000	101.090	47.846	-14.395	-956.114	0.000	917.789				
Max Mx	3	3	0.000	106.165	-2.846	-14.395	796.567	509.616	695.800				
Min Mx	1	3	0.000	101.090	47.846	-14.395	-956.114	0.000	917.789				

Beam Force Detail Summary Cont...

	Beam	L/C	d (m)	Axial Fx (kN)	Shear Fy (kN)	Fz (kN)	Torsion Mx (kip-in)	Bending My (kip-in)	Mz (kip-in)
Max My	3	3	0.000	106.165	-2.846	-14.395	796.567	509.616	695.800
Min My	2	3	5.000	-2.846	-106.165	-14.395	509.616	-796.567	695.800
Max Mz	1	3	0.000	101.090	47.846	-14.395	-956.114	0.000	917.789
Min Mz	2	3	2.500	-2.846	-15.261	-14.395	509.616	-478.057	-647.589

Beam Max Moments

Beam	Node A	Length (m)	L/C		d (m)	Max My (kip-in)	d (m)	Max Mz (kip-in)
1	1	4.000	1	Max + ve	0.000	0.000	4.000	46.252
				Max -ve	0.000	0.000	0.000	-10.588
			2	Max + ve	0.000	0.000	0.000	622.448
				Max -ve	4.000	-339.744	4.000	-32.618
			3	Max + ve	0.000	0.000	0.000	917.789
				Max -ve	4.000	-509.616	0.000	0.000
2	3	5.000	1	Max + ve	0.000	0.000	5.000	56.840
				Max -ve	0.000	0.000	2.500	-65.756
			2	Max + ve	0.000	0.000	5.000	407.026
				Max -ve	5.000	-531.045	2.083	-387.241
			3	Max + ve	0.000	0.000	5.000	695.800
				Max -ve	5.000	-796.567	2.083	-675.932
3	4	4.000	1	Max + ve	0.000	0.000	0.000	56.840
				Max -ve	0.000	0.000	4.000	0.000
			2	Max + ve	0.000	339.744	2.000	469.037
				Max -ve	4.000	0.000	0.000	0.000
			3	Max + ve	0.000	509.616	2.000	746.185
				Max -ve	4.000	0.000	0.000	0.000

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Beam Combined Stresses Summary

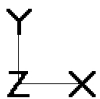
Beam	L/C	Length (m)	Max Comp			Max Tens		
			Stress (psi)	d (m)	Corner	Stress (psi)	d (m)	Corner
1	1	4.000	50.458	4.000	3	-33.757	4.000	1
	2	4.000	680.584	4.000	1	- 616.017	4.000	2
	3	4.000	1,007.4 77	4.000	1	- 885.575	4.000	2
2	1	5.000	61.158	2.500	1	-58.571	2.500	3
	2	5.000	1,334.6 57	5.000	4	- 1,340.3 02	5.000	2
	3	5.000	2,081.5 47	5.000	4	- 2,086.1 34	5.000	2
3	1	4.000	60.483	0.000	3	-43.011	0.000	1
	2	4.000	1,037.4 54	0.000	2	- 940.867	0.000	1
	3	4.000	1,646.9 06	0.000	2	- 1,475.8 17	0.000	1

Node Displacement Summary

Type	Node	L/C	X (in)	Y (in)	Z (in)	Result ant (in)	rX (rad)	rY (rad)	rZ (rad)
Max X	4	3	0.130	-0.005	-2.350	2.354	-0.012	0.017	0.000
Min X	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Max Y	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Min Y	4	3	0.130	-0.005	-2.350	2.354	-0.012	0.017	0.000
Max Z	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Min Z	4	3	0.130	-0.005	-2.350	2.354	-0.012	0.017	0.000
Max rX	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Min rX	2	3	0.000	0.000	0.000	0.000	-0.016	0.022	-0.002
Max rY	2	3	0.000	0.000	0.000	0.000	-0.016	0.022	-0.002
Min rY	1	1	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Max rZ	4	3	0.130	-0.005	-2.350	2.354	-0.012	0.017	0.000
Min rZ	2	3	0.000	0.000	0.000	0.000	-0.016	0.022	-0.002
Max Rst	4	3	0.130	-0.005	-2.350	2.354	-0.012	0.017	0.000

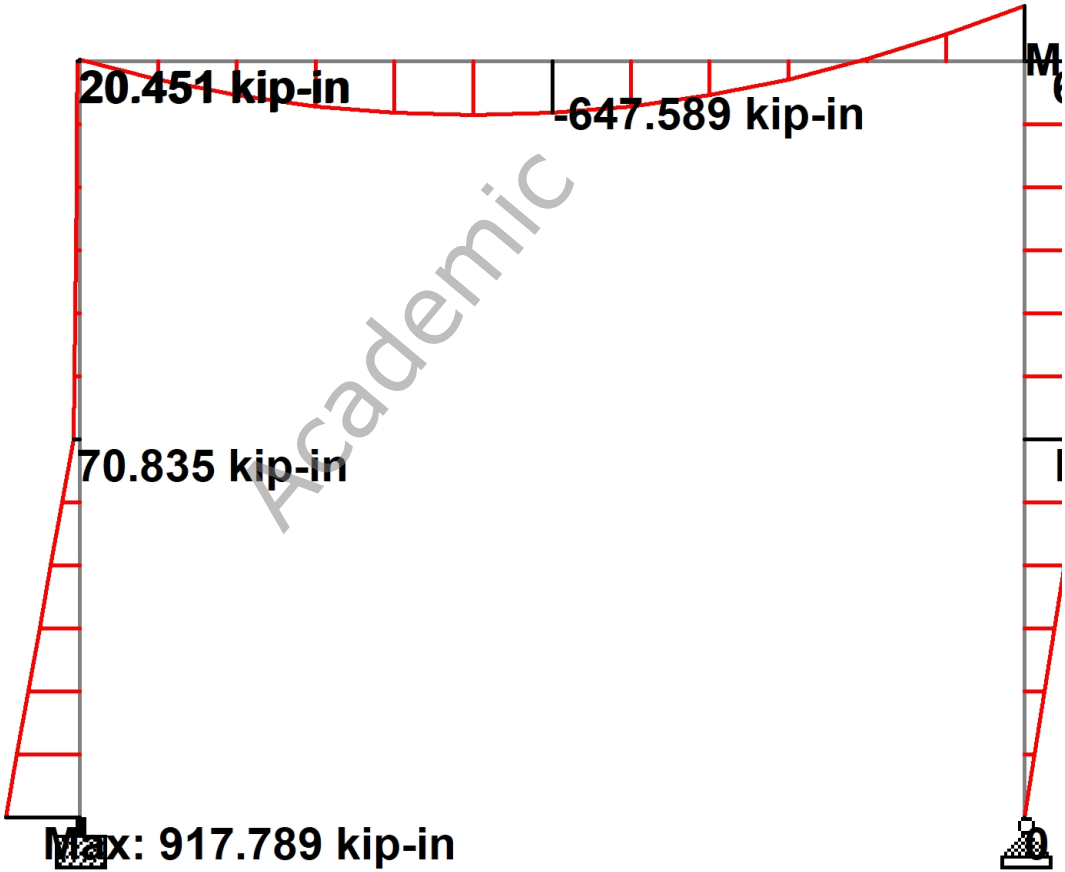
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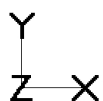
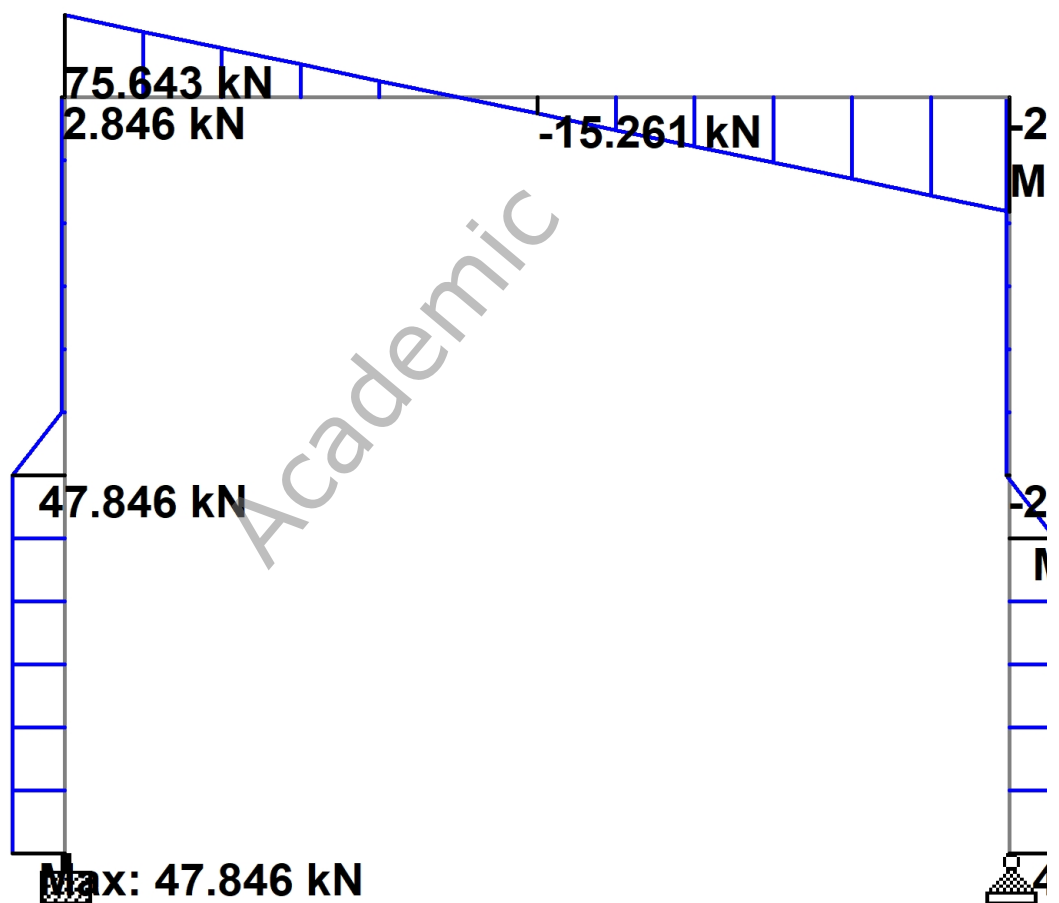
Load 3 : D
Disl

Whole Structure Displacements 0.23622in:1m 3 COMBINATION LOAD CASE 3



Load 3
M

Whole Structure Mz 2362.2kip-in:1m 3 COMBINATION LOAD CASE 3



Load

Whole Structure Fy 175.127kN:1m 3 COMBINATION LOAD CASE 3